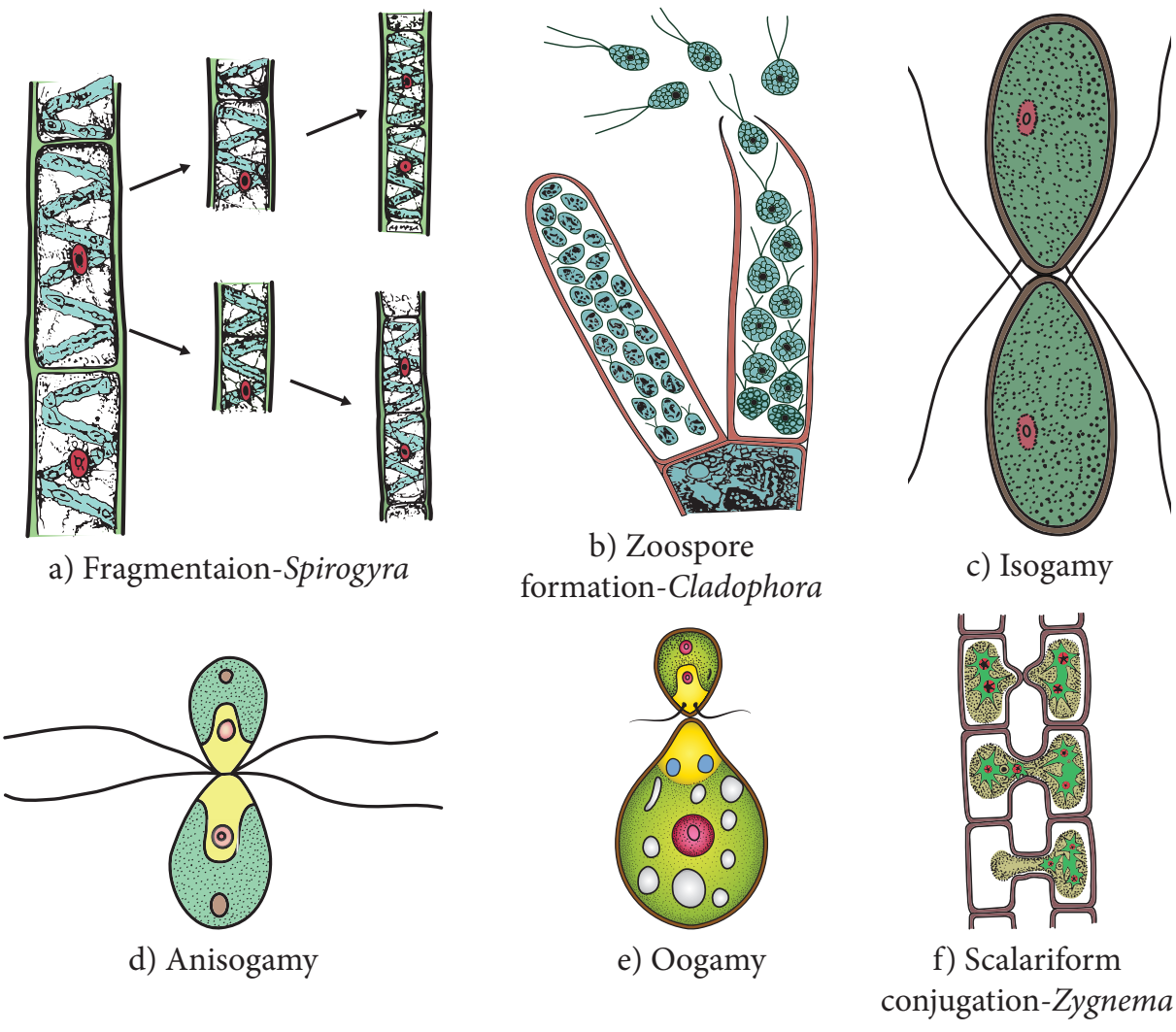
www.tntextbooks.in

sources for the alginate, agar agar and Carrageenan. The cell has a membrane bound nucleus and cell organelles like chloroplast, mitochondria, endoplasmic reticulum, golgi bodies etc., Pyrenoids are present. They are proteinaceous bodies found in chromatophores and assist in the synthesis and storage of starch. The pigmentation, reserve food material and flagellation differ among the algal groups.

Algae reproduces by vegetative, asexual and sexual methods (Figure 2.4). Vegetative reproduction includes fission (In unicellular forms the cell divides mitotically to produce two daughter cells Example: *Chlamydomonas)*; Fragmentation (fragments of parent thallus grow into new individual Example: *Ulothrix*) Budding (A lateral bud is formed in some members like *Protosiphon* and helps in reproduction) Bulbils, (a wedge shaped modified branch develop in *Sphacelaria*) Akinetes (Thick walled spores meant for perennation and germinates with the advent of favourable condition Example: *Pithophora*). Tubers (Structures found on the rhizoids and the lower nodes of *Chara* which store food materials).

Asexual reproduction takesplace by the production of zoospores(*Ulothrix, Oedogonium*) aplanospore(thin walled non motile spores Example: *Vaucheria*);



a) Fragmentaion-*Spirogyra*

b) Zoospore formation-*Cladophora*

c) Isogamy

d) Anisogamy

e) Oogamy

f) Scalariform conjugation-*Zygnema*

Figure 2.4: Reproduction in Algae